

An exponential two-way deterministic state lower bound for one-way liveness

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Abstract

One-way liveness on h points accepts a word of binary relations when their ordered product is nonempty. For every $h \geq 2$, it has a nondeterministic automaton with $h + 3$ states and no left moves, whereas every equivalent s -state two-way deterministic automaton satisfies $4(s + 2)^2 \geq 2^{\lfloor (h-2)/3 \rfloor}$. Partial transition rules, stay moves, and nonaccepting infinite computations are allowed. The alphabets are finite and grow with h , so the result rules out an alphabet-independent polynomial state bound for deterministic two-way simulation, already for one-way nondeterministic sources.

1 Introduction

Two-way deterministic automata recognize the same languages as one-way finite automata, as established by Rabin and Scott and by Shepherdson [13, 15]. Allowing the head to return to a symbol can nevertheless change the amount of finite control needed to recognize a language. How much finite control is needed to replace nondeterministic choices by deterministic head motion? Sakoda and Sipser posed the state cost of converting one-way and two-way nondeterministic finite automata to two-way deterministic automata in 1978 [14]. The question concerns succinctness: the simulator may revisit the input, but must store all information other than the head position in its finite state. A polynomial simulation would make nondeterminism inexpensive in this measure even when the source machine itself moves in both directions.

The witness used here is Sakoda and Sipser’s complete family B_h , now called *one-way liveness*: each letter specifies a binary relation on an h -element set, and a word is live when the product of its letters is nonempty. Their Theorem 2.3 establishes this family’s completeness for one-way-nondeterministic to two-way-deterministic state conversion [14]. Our proof below gives the small source automaton and the context-separation property directly, with the conventions and exact state counts used here.

Earlier exponential lower bounds restrict the motion of the deterministic target. Sipser proved an exponential separation for sweeping automata, whose head changes direction only at an endmarker [17]. Kapoutsis extended this phenomenon to deterministic machines constrained to make sublinearly many reversals as a function of input length [8]. An unrestricted target can instead reverse at an interior cell and make arbitrarily many reversals as the input length grows.

Quadratic lower bounds against unrestricted two-way deterministic targets already occur in Chrobak’s work on unary languages [5, 6]. For the promise restriction of one-way liveness to words of exactly three relation letters, Kapoutsis proved the optimal order $\Theta(h^2/\log h)$ [9]; the alphabet there is still the full relation alphabet. Adeogun and Kapoutsis [1] prove a quadratic lower bound for unrestricted deterministic two-way automata against one-way liveness. We prove

an exponential bound for that unrestricted target model. The source automata use no left moves, whereas the lower bound permits stay moves, partial transition rules, and nonaccepting infinite computations in the target.

The main representation difficulty is to account for repeated visits to a cell while assigning each input symbol a fixed object independent of its neighbors. We use the Brauer diagram monoid, whose matchings underlie Brauer's algebras [4, 3]: its elements are perfect matchings on two equally sized sets of ports, and multiplication glues the adjoining sets and contracts paths. Determinism makes the configurations reaching a designated accepting sink form a tree. A tour of that tree gives the required matching representation, with separate leaves for unused transition candidates ensuring that each cell's wiring depends only on its own symbol. The algebraic part then proves an exponential lower bound on the number of ports whenever matching products retain all the information in arbitrary relation products. All diagram and relation-monoid facts needed for this argument are proved below.

1.1 Model and statement

An automaton has a finite state set and one read-only head on a finite word over an alphabet Σ , bracketed by two distinct endmarkers outside Σ . Its head starts at the left endmarker in a designated initial state. A transition changes the state and moves the head left, right, or not at all; the head cannot cross an endmarker. A transition depends only on the current state and scanned symbol. Deterministic transition rules may be partial. A nondeterministic rule permits several successors. There are no additional heads, writable cells, work tapes, advice, or randomness. All states, including initial and accepting states, are counted.

Acceptance means that a finite computation reaches an accepting state; a nonaccepting infinite computation rejects. We initially count an accepting initial configuration as success. If success instead requires a positive transition into an accepting state, the normalization in Section 4 uses one additional initial copy. The main bound below includes this extra state and holds with either convention.

For a finite set H , let $\mathcal{R}_H$ be the monoid of all binary relations on H . Products are in path order from left to right:

$$(x, z) \in AB \iff \text{there exists } y \in H \text{ with } (x, y) \in A, (y, z) \in B.$$

Write $I_F = \{(x, x) : x \in F\}$ for $F \subseteq H$; the identity is I_H and the zero is the empty relation 0. For $H = [h]$ define

$$\text{OWL}_h = \{R_1 \cdots R_\ell \in (\mathcal{R}_{[h]})^* : R_1 \cdots R_\ell \neq 0\}. \quad (1.1)$$

The product of the empty word is $I_{[h]}$, so $\varepsilon \in \text{OWL}_h$.

Theorem 1.1. *For every integer $h \geq 2$, the language OWL_h is recognized by a nondeterministic automaton with $h+3$ states using no left moves. Every equivalent deterministic two-way automaton with s states satisfies*

$$4(s+2)^2 \geq 2^{\lfloor (h-2)/31 \rfloor}. \quad (1.2)$$

If an initially accepting configuration counts as acceptance, $s+2$ in (1.2) may be replaced by $s+1$.

Corollary 1.2. *There are no absolute constants $C, c > 0$ such that every n -state two-way nondeterministic automaton over every finite alphabet has an equivalent two-way deterministic automaton with at most Cn^c states.*

Proof. Put $n = h + 3$ in Theorem 1.1. Every deterministic equivalent for this family has at least

$$2^{\frac{1}{2}[(n-5)/31]-1} - 2$$

states. This bound exceeds Cn^c for every fixed C, c when n is large enough. For each h the alphabet has 2^{h^2} symbols and is finite. Since the constants in the proposed simulation must be independent of the alphabet, the family is within its scope. $\square$

1.2 Proof architecture

Write $\mathcal{D}_k$ for the matching monoid of degree k . The proof compares two bounds on k . Theorem 4.1 represents each s -state deterministic machine by symbol diagrams in $\mathcal{D}_k$, with $k \leq 4(s + 2)^2$. For a machine recognizing OWL_h , Lemma 5.2 gives a unital surjection from the submonoid generated by its ordinary-symbol diagrams onto $\mathcal{R}_{[h]}$. Concretely, the map sends each word diagram to its relation product, preserves multiplication and the empty product, and reaches every relation on $[h]$. These three properties, including the full image, are the algebraic input to the lower bound. Corollary 3.2 then forces $k \geq 2^{\lfloor (h-2)/31 \rfloor}$. These three statements give the deterministic lower bound in Theorem 1.1.

We prove the algebraic bound first. The *rank* of a matching diagram counts its pairs joining opposite sides. A diagram e is *idempotent* if $e^2 = e$; its *corner* consists of the diagrams z with $ez = ze = z$, with identity e . Section 2 reduces the corner of an idempotent of rank r to degree r , preserving rank and controlling the indices at which the diagram differs from the identity. An idempotent of rank r in $\mathcal{D}_k$ differs from the identity on at most $2(k - r)$ indices. The support control is uniform: a family supported in one set remains supported in a single set of no greater size after reduction. Section 3 uses 32 conjugates of one idempotent to make 256 successive relation additions. A common support set bounds their total rank loss, while each addition yields a further corner representing all relations on $h - 31$ points. Comparing these bounds doubles the required rank loss every 31 points.

Section 4 obtains local symbol diagrams by converting acceptance to reachability of a designated sink. Its undirected component is a tree, whose doubled tour supplies the matching test. This traversal has precedents in Sipser’s halting method and in the reversible simulation of Lange, McKenzie, and Tapp [16, 10]. Candidate transition slots completed by separate leaves make the construction independent of neighboring symbols. Finally, Section 5 obtains the quotient by using singleton-identity context letters to test each pair in a relation product. No input-length or running-time bound is imposed.

1.3 Related results

Adeogun and Kapoutsis also analyze limitations of their earlier lower-bound construction [2]. They prove a quadratic ceiling for its sequences of connectivity properties and state a general-property extension whose proof is deferred. The amplification here uses a fixed chain of 256 additions. Its exponential bound comes from the inductive lower bound on each rank loss in a smaller full relation corner, together with one common support budget for all additions.

The companion paper [11, Theorem 1.1] proves that a nondeterministic two-way automaton for the complement of OWL_h requires at least $\frac{1}{2}2^{\lfloor (h-2)/127 \rfloor} - 1$ states. The present proof establishes the deterministic bound directly using matching diagrams. Both results concern state succinctness over growing alphabets; neither yields a uniform logarithmic-space separation or a bound measured by transition-table size.

2 Matching and relation corners

We first establish the algebraic properties used in the rank argument. All monoid products in this paper are in path order: in a product ab , the factor a is placed to the left of the factor b . A homomorphism is called *unital* when it preserves the designated identities.

2.1 Matching diagrams, rank, and support

For an integer $m \geq 0$, let $[m] = \{1, \dots, m\}$, with $[0] = \emptyset$. A *matching diagram of degree m* is a perfect matching on two disjoint sets of ports, a left set and a right set, each indexed by $[m]$. Only the paired ports matter; crossings in a drawing have no significance. Let $\mathcal{D}_m$ be the set of these diagrams.

To multiply $a, b \in \mathcal{D}_m$, identify the right port indexed i of a with the left port indexed i of b , for every $i \in [m]$. In the resulting finite multigraph, each outside port has degree one and each identified port has degree two. Thus every component meeting an outside port is a path with two outside endpoints. Pair these endpoints and discard the components without outside ports. The result is ab . Parallel edges, when they occur at the interface, are counted separately.

This multiplication is associative. Indeed, in any string of glued diagrams, a substring can be replaced by the matching of its boundary endpoints: each path through the substring is replaced by one edge with the same endpoints. This preserves the connections between the outer ports of the whole string. Components internal to the substring have no connection to those ports. Contracting the first or the last two factors in a string of three therefore gives the same outside matching. The diagram $\mathbf{1}_m$ pairing each left port i with right port i is an identity. In particular, $\mathcal{D}_0 = \{\mathbf{1}_0\}$ is the one-element monoid. The monoids $\mathcal{D}_m$ are the Brauer diagram monoids, whose matchings underlie Brauer's algebras [4]. Their semigroup structure and transversal rank have been studied in particular by Auinger and by East and Gray [3, 7].

A pair joining opposite sides is a *transversal*; a pair on one side is a *cap*. Write $\text{rank}(a)$ for the number of transversals of a , and define

$$\text{supp}(a) = \{i \in [m] : a \text{ does not pair left } i \text{ with right } i\}.$$

We say that a is *supported in* $J \subseteq [m]$ if $\text{supp}(a) \subseteq J$.

Lemma 2.1 (Rank and support). *For $a, b \in \mathcal{D}_m$,*

$$\text{rank}(ab) \leq \min\{\text{rank}(a), \text{rank}(b)\}, \quad m - \text{rank}(a) \leq |\text{supp}(a)|.$$

The units of $\mathcal{D}_m$ are exactly its rank- m diagrams, which are the permutation diagrams. The only rank- m idempotent is $\mathbf{1}_m$. Conjugation by a permutation diagram preserves rank. For every $J \subseteq [m]$, the diagrams supported in J form a submonoid of $\mathcal{D}_m$.

Proof. Each transversal path in the product uses a transversal of a and a transversal of b . Different product paths are disjoint, so they use distinct transversals of either factor. This proves the rank inequality. Every index outside $\text{supp}(a)$ supplies a fixed transversal, proving the support inequality.

A rank- m diagram is a bijection between its two sides and has the inverse permutation diagram as a two-sided inverse. Conversely, the rank inequality forces every unit to have rank m . An idempotent in a group is its identity, proving the assertion about idempotents. If u is a permutation diagram, the rank inequality gives $\text{rank}(uau^{-1}) \leq \text{rank}(a)$; applying it to $a = u^{-1}(uau^{-1})u$ gives the reverse inequality.

If both a and b fix the transversal at an index i , their product also fixes it. Hence any product of diagrams supported in J is supported in J , and $\mathbf{1}_m$ is supported in every such set. $\square$

2.2 Reduction at an idempotent

If $e^2 = e$ in a monoid and S is a subset closed under multiplication and containing e , its *corner*

$$eSe = \{ese : s \in S\}$$

is a monoid with identity e . In particular, every $z \in eSe$ satisfies $ez = ze = z$. This is the standard local-monoid construction of finite-semigroup theory [12, Section 2.7]. For $m \geq 3$, Auinger observes that the corner of the standard rank- $(m-2)$ projection in $\mathcal{D}_m$ is isomorphic to $\mathcal{D}_{m-2}$ [3, Section 3]. The following reduction treats an arbitrary idempotent and realizes its matching corner on fewer ports. Its third part controls one common support set for an entire family of sandwiches.

Lemma 2.2 (Matching corners). *Let $e \in \mathcal{D}_m$ be idempotent and let $r = \text{rank}(e)$.*

(i) *There is a unital injective homomorphism*

$$R_e : e\mathcal{D}_m e \longrightarrow \mathcal{D}_r$$

satisfying $\text{rank}(R_e(z)) = \text{rank}(z)$ for all $z \in e\mathcal{D}_m e$.

(ii) *The idempotent satisfies*

$$|\text{supp}(e)| \leq 2(m-r).$$

(iii) *For every $J \subseteq [m]$ there is a set $J' \subseteq [r]$, depending only on e , J , and the numbering chosen for R_e , such that $|J'| \leq |J|$ and*

$$\text{supp}(R_e(eae)) \subseteq J' \quad \text{for every } a \in \mathcal{D}_m \text{ supported in } J.$$

Proof. Let $X, Y \subseteq [m]$ be the indices of the left and right transversal endpoints of e , respectively. Its transversals define a bijection $p : X \rightarrow Y$, with $|X| = |Y| = r$.

We describe the connector at the interface between two copies of e . Keep two distinct tagged rows of vertices, R_i and L_i for $i \in [m]$, and put an edge $R_i L_i$ between each matching pair of indices. On the R row put the right caps of e , and on the L row put the left caps of e . Thus the degree-one vertices are precisely

$$\{R_y : y \in Y\} \cup \{L_x : x \in X\},$$

and all other vertices have degree two. These tagged vertices remain distinct even when X and Y overlap. Pairing the endpoints of its paths, and discarding its closed components, gives a matching on the $2r$ displayed vertices.

This is exactly the connector encountered by the transversals in $e\mathbf{1}_m e = e^2$. Since $\text{rank}(e^2) = r$, all r left transversal endpoints of the first copy must reach the right boundary of the second. Consequently the connector cannot pair two R endpoints or two L endpoints: it pairs R_y with $L_{\gamma(y)}$ for a bijection $\gamma : Y \rightarrow X$. The path from an outer left endpoint $x \in X$ ends at the right endpoint $p(\gamma(p(x)))$. Equality $e^2 = e$ gives

$$p(\gamma(p(x))) = p(x), \quad \text{hence} \quad \gamma(p(x)) = x.$$

Thus $\gamma = p^{-1}$.

Every diagram $z \in e\mathcal{D}_m e$ contains the left caps of e on its outer left side and the right caps of e on its outer right side. Delete precisely these prescribed caps. The remaining ports are the

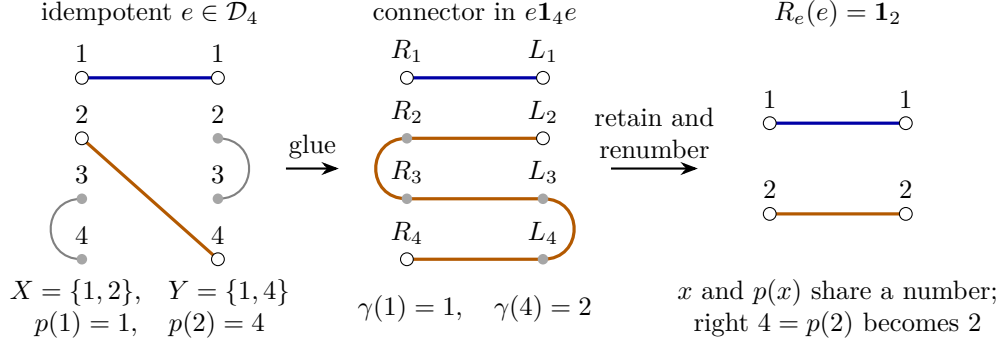


Figure 1: A rank-2 idempotent of degree 4 and its reduction. In the left panel, the gray caps are prescribed and the colored transversals are retained. The connector paths are R_1-L_1 and $R_4-L_4-L_3-R_3-R_2-L_2$, giving $\gamma = p^{-1}$. Numbering left x and right $p(x)$ alike yields $R_e(e) = \mathbf{1}_2$. For a general corner element z , only the prescribed outer caps are deleted; every pair among retained ports, including caps, remains.

left ports indexed by X and the right ports indexed by Y , and they retain a perfect matching. In particular, any additional caps among these retained ports remain in the reduced diagram. Number X by $[r]$ in any fixed way, and give the right endpoint $p(x)$ the same number as x . This defines $R_e(z)$.

Figure 1 illustrates the connector and the numbering for a rank-2 idempotent.

Restoring the prescribed caps and the original indices reconstructs z , so R_e is injective. Only nontransversal pairs were deleted, so it preserves rank, and by construction $R_e(e) = \mathbf{1}_r$. To check multiplication, let $z, w \in e\mathcal{D}_m e$. The fixed caps at their middle interface are the right caps and left caps of e . Their connector is therefore the one just computed: the retained right port $p(x)$ of z is connected to the retained left port x of w . In the reduced numbering these ports have the same index. Contracting the connector, including when the reduced diagrams have caps of their own, thus gives ordinary multiplication in $\mathcal{D}_r$. It follows that $R_e(zw) = R_e(z)R_e(w)$, proving (i).

For (ii), suppose that $i \in X \cap Y$. The vertices R_i, L_i in the connector have no incident caps, so their edge $R_i L_i$ is an entire connector path. Hence $\gamma(i) = i$, and $\gamma = p^{-1}$ implies $p(i) = i$. Thus every index in $X \cap Y$ is fixed by e , and

$$\text{supp}(e) \subseteq ([m] \setminus X) \cup ([m] \setminus Y).$$

Both complements have size $m - r$, which proves the bound.

For (iii), consider the full glued graph $e\mathbf{1}_m e$, keeping the middle identity diagram explicit. Its transversal paths have the same outside endpoint pairs as e , and so are indexed by the r numbers used in the reduction. Mark a transversal path if it uses any middle identity edge whose index belongs to J . Let J' be the numbers of the marked paths. A transversal path may use several middle identity edges; the only counting fact needed is that different paths are disjoint. Each middle identity edge therefore marks at most one transversal path, and $|J'| \leq |J|$.

Now replace the middle identity by any diagram a supported in J . Every middle identity edge at an index outside J remains present. An unmarked transversal path used only such edges, so that entire path survives with the same outside endpoints. Each of its internal vertices still has its two original incident edges; a new path cannot attach to it. In particular, changes to closed components or to paths with both endpoints on the same side cannot join an unmarked transversal: all its middle ports are already paired by the unchanged identity edges. Its corresponding reduced

port is therefore still paired with the equally indexed port on the other side. Only numbers in J' can belong to $\text{supp}(R_e(eae))$. The construction of J' did not depend on a , as required.

When $r = 0$, the prescribed caps exhaust all outside ports. Then $eae = e$ for every a , and the corner is identified with the singleton $\mathcal{D}_0$. The reduction above is its unique map, and (iii) holds with $J' = \emptyset$; (ii) also holds since $|\text{supp}(e)| \leq m$. $\square$

The reduction will also be applied to a submonoid eSe . We write $R_e(eSe)$ for its image in $\mathcal{D}_r$. It contains the ambient identity 1_r . When discussing support in this reduced monoid, the port set is $[r]$; support in the original monoid still refers to $[m]$.

2.3 Homomorphic images and permutation lifts

Every element of a finite semigroup has an idempotent positive power. Indeed, for an element x there are integers $a \geq 1$ and $d \geq 1$ such that $x^{k+d} = x^k$ for all $k \geq a$. Choose $N \geq a$ divisible by d . Then $x^{2N} = x^N$. If a multiplicative map sends x to an idempotent q , the same is true of every positive power of x , including this idempotent power.

Lemma 2.3 (Unit lifts from a minimum-rank corner). *Let $S \subseteq \mathcal{D}_m$ be nonempty and closed under multiplication, and let $\phi : S \rightarrow M$ be an onto multiplicative map to a monoid M . Let $q \in M$ be idempotent. There is an idempotent $e \in S$ with $\phi(e) = q$. For every such e , the reduced monoid $R_e(eSe)$ has a unital surjection*

$$\phi_e : R_e(eSe) \longrightarrow qMq, \quad \phi_e(R_e(z)) = \phi(z).$$

If e has minimum rank among all idempotents in S mapping to q , then every element of eSe mapping to a unit of qMq reduces to a permutation diagram. In particular, every unit of qMq has a permutation lift in $R_e(eSe)$, and that lift is a unit of the reduced monoid.

Proof. The sets S and M are finite because $\mathcal{D}_m$ is finite and ϕ is onto. Take any preimage of q and then an idempotent positive power to obtain e . The image of eSe under ϕ is exactly qMq : for any $v \in M$, a preimage $s \in S$ gives $\phi(ese) = qvq$. Its identity e maps to the identity q of this corner. Lemma 2.2(i) makes the displayed definition of ϕ_e well-defined and transfers both multiplication and surjectivity.

Suppose now that e has the stated minimum rank, put $r = \text{rank}(e)$, and let $y \in eSe$ map to a unit u of qMq . The group of units of qMq is finite, so $u^t = q$ for some positive integer t . Take an idempotent positive power z of y^t . Then $z \in S$ is idempotent and $\phi(z) = q$. The rank inequalities and minimum choice of e give

$$r \leq \text{rank}(z) \leq \text{rank}(y) \leq r.$$

The last inequality uses $y \in eSe$. Hence $R_e(y)$ has rank r in $\mathcal{D}_r$ and is a permutation diagram. Its inverse is a power of the same finite-order permutation, so lies in $R_e(eSe)$. Surjectivity of ϕ_e supplies a lift for each unit u and completes the proof. $\square$

2.4 Corners of the relation monoid

For a finite set H and a subset $F \subseteq H$, we regard a relation on F as a relation on H containing no pairs outside $F \times F$. With $i = I_F$, these are precisely the relations A satisfying $A = iAi$. Under this identification their monoid is $i\mathcal{R}_H i$ with identity i . Permutations of a finite set are identified with their graph relations.

Lemma 2.4 (A relation corner). *Let $P \in \mathcal{R}_H$ be idempotent, and let $F \subseteq H$ satisfy $I_F P I_F = I_F$. Put $i = I_F$ and $Q = P i P$. Then Q is idempotent, $PQ = QP = Q$, and restriction is a unital isomorphism*

$$\theta : Q\mathcal{R}_H Q \longrightarrow \mathcal{R}_F, \quad \theta(Z) = iZi.$$

Its inverse sends $A \in \mathcal{R}_F$ to $PAP = QAQ$.

Proof. The identities $P^2 = P$ and $iPi = i$ give

$$Q^2 = P i P^2 i P = P(iPi)P = Q, \quad PQ = QP = Q.$$

They also give $iQ = iP$, $Qi = Pi$, and $iQi = i$. If $Z \in Q\mathcal{R}_H Q$, then $Z = QZQ$ and $PZP = Z$. Consequently

$$Z = P i P Z P i P = P i Z i P. \quad (*)$$

Thus its restriction iZi determines Z uniquely.

Conversely, if $A \in \mathcal{R}_F$, so that $A = iAi$, then

$$QAQ = Q i A i Q = P i A i P = PAP, \quad i(PAP)i = i P i A i P i = A.$$

Hence every relation on F is a restriction of an element of the corner, and $A \mapsto PAP$ is the inverse bijection by (*).

For multiplication, write $Z = PAP$ and $W = PBP$, where $A = iZi$ and $B = iWi$ are relations on F . Since $A = Ai$ and $B = iB$, we have

$$ZW = PAP^2BP = PAPBP = PA(iPi)BP = PABP.$$

Restricting this equality gives $\theta(ZW) = AB = \theta(Z)\theta(W)$. Finally, the identity Q of the source corner restricts to i , the identity of $\mathcal{R}_F$. This proves unitality and the claimed isomorphism. $\square$

3 An exponential rank-loss bound

We now bound the rank lost by an idempotent diagram whose image adds one off-diagonal pair to the identity relation. The amplification step uses 32 conjugates of that diagram to construct 256 successive relation additions. Their joint support bounds the total rank loss, while a smaller relation corner bounds the loss at each addition.

For every integer $h \geq 2$, set

$$L(h) = 2^{\lfloor (h-2)/31 \rfloor}. \quad (3.1)$$

All reductions below are the maps R_e of Lemma 2.2. Their ranges are regarded as submonoids of the indicated smaller matching monoids. In particular, their identities are the ambient identity diagrams, and they preserve absolute rank.

The theorem first assumes permutation lifts in addition to a unital surjection onto the *entire* relation monoid. A surjection onto only the relations generated by the displayed additions would not suffice: the proof must also lift arbitrary elements of smaller relation corners. Corollary 3.2 removes the extra lift assumption by passing to a minimum-rank idempotent corner.

Theorem 3.1 (Rank loss for a single relation addition). *Let H be a finite set of size $h \geq 2$, let $m \geq 0$, and let $S \subseteq \mathcal{D}_m$ be a submonoid containing $\mathbf{1}$. Suppose that $\phi : S \rightarrow \mathcal{R}_H$ is a unital*

surjection and that every permutation of H has a lift in S which is a permutation diagram. If $a \in S$ is idempotent and

$$\phi(a) = I_H \cup \{(x, y)\}, \quad x, y \in H, \quad x \neq y,$$

then

$$m - \text{rank}(a) \geq L(h).$$

Proof. We induct on h , simultaneously for every ambient degree m and every choice of S, ϕ, a satisfying the hypotheses. Put $t = 16$. For $2 \leq h < 2t + 1 = 33$, one has $L(h) = 1$. A full-rank diagram is a permutation, and an idempotent permutation is the identity. Since ϕ is unital and $\phi(a) \neq I_H$, the rank of a is strictly smaller than m . This proves the base cases.

Now let $h \geq 33$ and assume the theorem for every smaller size at least two. Write

$$c = m - \text{rank}(a).$$

Conjugates with a common support bound. Choose pairwise disjoint sets

$$U = \{u_1, \dots, u_t\}, \quad V = \{v_1, \dots, v_t\}, \quad \{z\}$$

in H . For each ordered pair of distinct points of H , a permutation π can send (x, y) to that pair. Choose a permutation-diagram lift $\tilde{\pi} \in S$. Its inverse belongs to S because a permutation diagram has finite order. In path order,

$$\phi(\tilde{\pi}^{-1} a \tilde{\pi}) = \pi^{-1}(I_H \cup \{(x, y)\})\pi = I_H \cup \{(\pi(x), \pi(y))\}.$$

Using the pairs (u_i, z) and (z, v_j) in this construction gives idempotents $s_i, s'_j \in S$ such that

$$\phi(s_i) = I_H \cup \{(u_i, z)\}, \quad \phi(s'_j) = I_H \cup \{(z, v_j)\} \quad (1 \leq i, j \leq t). \quad (3.2)$$

Conjugation by permutation diagrams preserves rank, so all these idempotents have rank $m - c$. By Lemma 2.2(ii), each has support of size at most $2c$. Thus the set

$$J = \bigcup_{i=1}^t \text{supp}(s_i) \cup \bigcup_{j=1}^t \text{supp}(s'_j) \subseteq [m]$$

has size at most $4tc$. Every product $s_i s'_j$ is supported in J , because diagrams supported in a fixed set form a submonoid.

A chain of t^2 additions. Put $P_0 = I_{H \setminus \{z\}}$. Choose an idempotent $b_0 \in S$ with $\phi(b_0) = P_0$: take any lift of P_0 and then a positive idempotent power. The image stays P_0 because P_0 is idempotent. For $1 \leq i, j \leq t$, define

$$g_{ij} = b_0 s_i s'_j b_0 \in b_0 S b_0.$$

In path order, the two added pairs in (3.2) allow the path u_i, z, v_j . The outer factors P_0 delete pairs with endpoint z , so

$$\phi(g_{ij}) = P_0 \cup \{(u_i, v_j)\}. \quad (3.3)$$

Figure 2 shows how the same $2t$ conjugates provide all t^2 choices of the pair (u_i, v_j) .

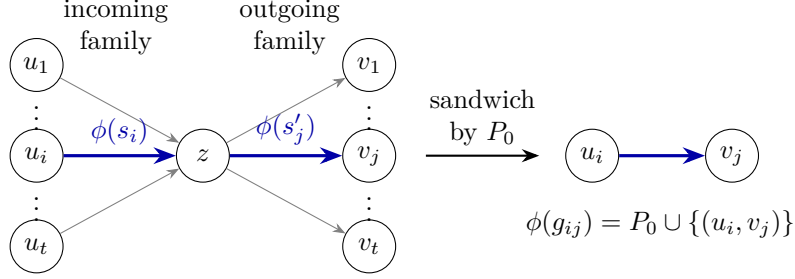


Figure 2: The gray arrows are alternatives from the two families of relation images; each individual image has one added arrow. The highlighted path creates the selected pair after both outer factors delete endpoints at z . Identity loops are omitted. Only relations on H are shown. The common matching support bound comes from taking the union of the supports of the 32 conjugates and reducing at b_0 ; the same conjugates provide all 256 choices.

Let $r_0 = \text{rank}(b_0)$. Applying Lemma 2.2(iii) at b_0 to the single set J gives one set $J_0 \subseteq [r_0]$, with $|J_0| \leq 4tc$, such that

$$\text{supp}(R_{b_0}(g_{ij})) \subseteq J_0 \quad (1 \leq i, j \leq t). \quad (3.4)$$

This set is common to all t^2 generators.

List the pairs $(i, j) \in [t] \times [t]$ in any order, without repetition. Let E_ℓ be the set of relation pairs (u_i, v_j) in the first ℓ positions, and set

$$P_\ell = P_0 \cup E_\ell \quad (0 \leq \ell \leq t^2).$$

All additional pairs go from U to V , and $U \cap V = \emptyset$. Consequently two additional pairs cannot be composed with each other; the identity pairs in P_0 preserve their endpoints. Thus

$$(P_0 \cup A)(P_0 \cup B) = P_0 \cup (A \cup B) \quad (A, B \subseteq U \times V). \quad (3.5)$$

In particular, $P_\ell^2 = P_\ell$.

Starting from b_0 , construct b_ℓ as follows. If (i, j) occupies position ℓ , take b_ℓ to be a positive idempotent power of

$$b_{\ell-1}g_{ij}b_{\ell-1}.$$

Inductively, its image before taking the power is P_ℓ , by (3.3) and (3.5); the power has the same image. Hence

$$b_\ell^2 = b_\ell, \quad b_\ell \in b_{\ell-1}Sb_{\ell-1}, \quad \phi(b_\ell) = P_\ell. \quad (3.6)$$

Every b_ℓ also belongs to b_0Sb_0 . Indeed, this is true initially, and this latter set is a submonoid containing every g_{ij} ; it is therefore preserved by the construction.

Write

$$r_\ell = \text{rank}(b_\ell) \quad (0 \leq \ell \leq t^2), \quad \Delta_\ell = r_{\ell-1} - r_\ell \quad (1 \leq \ell \leq t^2).$$

The nested corners in (3.6) and Lemma 2.1 give $\Delta_\ell \geq 0$. In the single coordinate system obtained by reducing at b_0 , the starting element is the identity and all generators are supported in J_0 by (3.4). The same is true of every constructed product and power, in particular $R_{b_0}(b_{t^2})$. Rank preservation and the elementary bound of rank defect by support size give

$$\sum_{\ell=1}^{t^2} \Delta_\ell = r_0 - r_{t^2} \leq |\text{supp}(R_{b_0}(b_{t^2}))| \leq |J_0| \leq 4tc. \quad (3.7)$$

The loss at one addition. We claim that for every $1 \leq \ell \leq t^2$,

$$2\Delta_\ell \geq L(h - 2t + 1). \quad (3.8)$$

Each local argument uses fresh port coordinates to certify this bound on the already defined absolute rank difference Δ_ℓ in (3.7). Fix ℓ , let (i, j) be its pair, and put $P = P_{\ell-1}$. Reduce at $b_{\ell-1}$ and define

$$T = R_{b_{\ell-1}}(b_{\ell-1} S b_{\ell-1}) \subseteq \mathcal{D}_{r_{\ell-1}}, \quad \beta = R_{b_{\ell-1}}(b_\ell) \in T.$$

Here T contains the ambient identity $R_{b_{\ell-1}}(b_{\ell-1}) = \mathbf{1}$. The unital-surjection assertion of Lemma 2.3 gives

$$\theta : T \longrightarrow P\mathcal{R}_H P, \quad \theta(R_{b_{\ell-1}}(w)) = \phi(w) \quad (w \in b_{\ell-1} S b_{\ell-1}).$$

Moreover, β is idempotent, $\theta(\beta) = P_\ell$, and rank preservation gives

$$\text{rank}(\beta) = r_\ell, \quad |\text{supp}(\beta)| \leq 2\Delta_\ell, \quad (3.9)$$

where support now refers to the indices $[r_{\ell-1}]$.

A full relation corner on fewer points. Set

$$F = \{u_i, v_j\} \cup (H \setminus (U \cup V \cup \{z\})), \quad E = I_F, \quad Q = PEP.$$

Thus $|F| = h - 2t + 1 = h - 31$. Among the possible extra pairs in P , the only one with both endpoints in F is (u_i, v_j) , which has not yet been added. Every point of F is different from z . Therefore

$$EPE = E.$$

Lemma 2.4 shows that Q is idempotent, $PQ = QP = Q$, and restriction $Z \mapsto EZE$ is a unital isomorphism from $Q\mathcal{R}_H Q$ onto $\mathcal{R}_F$. In particular,

$$Q(P\mathcal{R}_H P)Q = Q\mathcal{R}_H Q.$$

Equation (3.5) also gives $PP_\ell = P_\ell P = P_\ell$. Since $EQ = EP$ and $QE = PE$, we obtain

$$E(QP_\ell Q)E = EPP_\ell PE = EP_\ell E = I_F \cup \{(u_i, v_j)\}. \quad (3.10)$$

The second reduction and the induction hypotheses. Among the idempotents $d \in T$ with $\theta(d) = Q$, choose one of minimum rank. Such idempotents exist because θ is onto and an idempotent power of a lift of Q still maps to Q . This minimum is taken inside T , in its current coordinates in $\mathcal{D}_{r_{\ell-1}}$. Put $r' = \text{rank}(d)$ and reduce once more, defining

$$T' = R_d(dTd) \subseteq \mathcal{D}_{r'}.$$

The identity of T' is $R_d(d) = \mathbf{1}$. Its induced map onto $Q(P\mathcal{R}_H P)Q$, followed by restriction to F , is the unital surjection

$$\psi : T' \longrightarrow \mathcal{R}_F, \quad \psi(R_d(w)) = E\theta(w)E \quad (w \in dTd).$$

By Lemma 2.3 and the minimum-rank choice of d , every unit of $Q(P\mathcal{R}_H P)Q$ has a permutation-diagram lift in T' . The restriction isomorphism preserves units, so every permutation of F has such a lift under ψ .

Let $J_\ell = \text{supp}(\beta) \subseteq [r_{\ell-1}]$. Apply Lemma 2.2(iii) to the idempotent d in this ambient matching monoid. There is a set $K \subseteq [r']$ with $|K| \leq |J_\ell| \leq 2\Delta_\ell$ such that

$$\text{supp}(R_d(d\beta d)) \subseteq K.$$

The sandwich $d\beta d$ need not be idempotent. Choose a positive idempotent power v of its reduction $R_d(d\beta d)$ in T' . Support remains in K under taking powers. By (3.10), the image before taking the power is $I_F \cup \{(u_i, v_j)\}$, an idempotent relation; hence

$$v^2 = v, \quad \psi(v) = I_F \cup \{(u_i, v_j)\}, \quad |\text{supp}(v)| \leq 2\Delta_\ell.$$

We have obtained an identity-containing submonoid of $\mathcal{D}_{r'}$, a unital surjection onto $\mathcal{R}_F$ with permutation lifts, and the required idempotent above a single off-diagonal addition. These are all the hypotheses of the theorem at size $|F| = h - 31$, which lies between 2 and $h - 1$. The induction hypothesis and the support bound therefore give

$$L(h - 31) \leq r' - \text{rank}(v) \leq |\text{supp}(v)| \leq 2\Delta_\ell.$$

This proves (3.8) for the chosen ℓ , and hence for every step of the chain.

Completion of the recurrence. The bounds established above are collected in Table 1. The support is measured once, in the coordinates of b_0 ; the separate smaller corners certify each summand in the resulting rank-loss budget.

Quantity	General expression	$t = 16$
Conjugates	$2t$	32
Successive additions	t^2	256
Common support budget	$4tc$	$64c$
Retained relation points	$h - 2t + 1$	$h - 31$
Loss per addition	$\frac{1}{2}L(h - 2t + 1)$	$\frac{1}{2}L(h - 31)$
Total required loss	$\frac{t^2}{2}L(h - 2t + 1)$	$128L(h - 31)$

Table 1: The support upper bound and the inductive rank-loss lower bound are applied to the same chain. Here $c = m - \text{rank}(a)$.

Summing (3.8) and applying (3.7) yields

$$4tc \geq \sum_{\ell=1}^{t^2} \Delta_\ell \geq \frac{t^2}{2}L(h - 2t + 1).$$

With $t = 16$, it follows that

$$c \geq 2L(h - 31) = 2^{1+\lfloor (h-33)/31 \rfloor} = 2^{\lfloor (h-2)/31 \rfloor} = L(h).$$

This completes the induction. $\square$

Corollary 3.2 (Degree bound for a unital relation-monoid divisor). *Let H be a finite set of size $h \geq 2$. If an identity-containing submonoid $S \subseteq \mathcal{D}_k$ admits a unital surjection $\phi : S \rightarrow \mathcal{R}_H$, then*

$$k \geq L(h) = 2^{\lfloor (h-2)/31 \rfloor}.$$

No hypothesis about permutation lifts is required here.

Proof. Choose an idempotent $e \in S$ of minimum rank among those with $\phi(e) = I_H$. The set of choices is nonempty, since the identity of S is one such idempotent. Put $r = \text{rank}(e)$. By Lemmas 2.2 and 2.3, the reduced corner

$$T = R_e(eSe) \subseteq \mathcal{D}_r$$

contains the ambient identity and maps unitally onto $I_H \mathcal{R}_H I_H = \mathcal{R}_H$. The minimum-rank choice supplies permutation-diagram lifts for all permutations of H .

Choose distinct $x, y \in H$. Surjectivity supplies a lift in T of $I_H \cup \{(x, y)\}$. A positive idempotent power of that lift is an idempotent $a \in T$ with the same image. Theorem 3.1 applies to this representation, so

$$L(h) \leq r - \text{rank}(a) \leq r \leq k. \quad \square$$

4 Local matching representation of deterministic machines

Fix a finite alphabet Σ . We use one read-only head on an input $\vdash w \dashv$, where $w \in \Sigma^*$ and the two endmarkers are distinct symbols outside Σ . A deterministic machine has a finite state set Q , an initial state q_0 , a set $A \subseteq Q$ of accepting states, and a partial transition function

$$\delta : Q \times (\Sigma \cup \{\vdash, \dashv\}) \rightarrow Q \times \{-1, 0, 1\}.$$

The three moves are left, stay, and right. A transition at an endmarker never moves beyond that endmarker. The initial configuration is $(0, q_0)$, with cell 0 carrying $\vdash$. In this section a computation accepts when an accepting state occurs at any time, including time 0. A missing transition halts the computation; an infinite computation without an accepting state rejects. All states, including initial and accepting states, count toward $s = |Q|$.

Theorem 4.1 (Local matching representation). *For every s -state deterministic machine in this model, there is a map*

$$\eta : \Sigma \cup \{\vdash, \dashv\} \longrightarrow \mathcal{D}_k, \quad k = 4(s+1)^2,$$

with the following property. Designate ports 1, 2 on each of the two sides as test ports. For every word $w = a_1 \cdots a_\ell$, the machine accepts w if and only if the diagram

$$\eta(\vdash)\eta(a_1) \cdots \eta(a_\ell)\eta(\dashv)$$

pairs a left test port with a right test port. Each symbol diagram is independent of the input length, the symbol's position, and its neighboring symbols.

If acceptance instead requires entry into an accepting state after a positive number of transitions, the same conclusion holds with $k = 4(s+2)^2$.

The diagrams will encode a tour of an undirected graph. Determinism is used to ensure that the component of one designated vertex is a tree; no condition is imposed on the other components. Configuration-tree traversal also underlies Sipser's halting method [16] and the reversible simulation of Lange, McKenzie, and Tapp [10, Section 3.1]. We prove the traversal facts needed here and then realize them by matching diagrams. The candidate-slot construction below supplies the further requirement that each diagram depend only on the symbol at its own cell, with the stated quadratic degree.

4.1 A sink and its tree

Lemma 4.2 (Sink component). *Let V be finite, let $\sigma : V \rightharpoonup V$ be a partial function, and let $t \in V$ have no successor. Make an undirected multigraph on V with one edge for each defined successor $v \mapsto \sigma(v)$, retaining self-loops and distinct edges with the same endpoints. Then a vertex lies in the connected component of t if and only if some finite sequence of successor steps takes it to t . This component is a tree.*

Proof. Let B be the set of vertices reaching t , allowing a sequence of length zero. For every successor edge $u \mapsto v$, one has $u \in B$ if and only if $v \in B$. One implication prepends that edge to a path to t ; the other removes the first step of the unique successor path from u to t . This removal is legitimate because $u \neq t$. Thus membership in B is constant along undirected edges. Since $t \in B$, its entire component lies in B . Conversely, a successor path is an undirected path, so every element of B belongs to that component.

If the component has v vertices, each of its vertices other than t has exactly one outgoing edge, and t has none. All these edges stay in the component, and every edge in the component is counted once at its source. Hence the connected multigraph has $v - 1$ edges and is a tree. In particular, it has neither self-loops nor parallel edges. $\square$

To apply the lemma, add one new state f to the machine and put $N = s + 1$. This is an auxiliary successor rule for a reachability test, not a change in the language to be recognized. For every $b \in \Sigma \cup \{\vdash, \dashv\}$, define

$$\hat{\delta}(q, b) = \begin{cases} (f, 0), & q \in A, \\ \delta(q, b), & q \in Q \setminus A, \\ (f, 1), & q = f, b \neq \dashv, \\ \text{undefined}, & q = f, b = \dashv. \end{cases}$$

The first case applies even when the old transition was undefined. Only the rows at old nonaccepting states retain their old partial rules, including undefined entries.

For a fixed input of length ℓ , let

$$c_0 = (0, q_0), \quad t = (\ell + 1, f).$$

The original machine accepts precisely when the modified successor rule reaches t from c_0 . Indeed, the computations agree until the first accepting state, which now leads into f and then to t . Conversely, the first entry into f must come from an old accepting state. This also covers acceptance initially or on an endmarker. Let G be the undirected successor multigraph on all configurations (i, q) , where $0 \leq i \leq \ell + 1$ and $q \in Q \cup \{f\}$. Lemma 4.2 shows that the component of t is a tree, and that c_0 belongs to it exactly on accepted inputs. Nonaccepting cycles in other components require no further modification.

4.2 Tours determined by cyclic orders

An incidence of an edge at a vertex records its endpoint there. A loop has two incidences at its vertex. Choose a cyclic order of the incidences at every vertex of positive degree. Regard each edge as having two directed traversals. Upon arriving at a vertex along one incidence, depart along the next incidence in its cyclic order. At a vertex of degree one this means returning along the same edge.

Lemma 4.3 (Tree tour). *For a finite tree with at least one edge, this rule has a single cycle on its directed edge traversals, for every choice of cyclic orders. If the two joins at two distinct leaves are opened, the resulting two paths each connect a tip at one leaf to a tip at the other.*

Proof. The rule is a permutation of the directed traversals: a departure has exactly one preceding arrival, determined by the preceding incidence in the cyclic order. A tree consisting of one edge gives a cycle of its two traversals.

For the induction step, remove a leaf z with neighbor v from a tree with more than one edge. Restrict the cyclic order at v by removing the incidence of vz . The smaller tree has one tour cycle by induction. Inserting vz back into that order replaces one transition

$$(u, v) \mapsto (v, w)$$

of the smaller tour by

$$(u, v) \mapsto (v, z) \mapsto (z, v) \mapsto (v, w).$$

Here $u = w$ is allowed. No other transition changes. Thus two new traversals are inserted into the existing cycle, proving the first assertion.

Realize the tour as a circular wire, using one lane for each directed traversal and a join for each arrival-to-departure step. The joins at two different leaves are two different points on this circle. Opening both cuts the circle into two paths, each running between the two cuts. Each path consequently has one tip at each of the two leaves. $\square$

4.3 Cell diagrams and their composition

Proof of Theorem 4.1. Use the modified successor rule with its $N = s + 1$ states, numbered by $[N]$. We first augment its undirected graph by leaves so that each cell's portion of the graph can be determined from that cell's symbol alone.

Candidate edges at an internal boundary. At every boundary between two neighboring cells, reserve one slot for each triple

$$(d, p, q) \in \{\leftarrow, \rightarrow\} \times [N] \times [N].$$

The direction d specifies which cell is the prospective source and which is the prospective target. The slot represents a candidate transition from state p in the source cell to state q in the target cell. It carries two lane ports, labeled by the absolute directions of travel $\rightarrow$ and $\leftarrow$. These labels, together with (d, p, q) , give a fixed enumeration of the

$$k = 2 \cdot N^2 \cdot 2 = 4N^2$$

ports on either side of a cell. Equal labels are identified when neighboring cell diagrams are multiplied.

For each slot put an undirected edge across the boundary. At its target end, always attach it to the target cell's state vertex q . At its source end, attach it to state vertex p if the modified transition from p on the source cell's symbol is exactly the transition specified by the slot. Otherwise attach that end to a new degree-one vertex in the source cell. Use a different new vertex for each unused slot.

Every actual crossing successor edge is represented once, in its own slot. Each other slot contributes a single leaf edge attached to a state vertex. Keep every actual stay edge inside its cell,

with two distinct incidences when it is a loop. The resulting graph is therefore G with additional leaves and leaf edges. It has the same connectivity between state vertices as G , and the component of t is still a tree.

The target-end attachment makes the construction local: a target cell does not need to know whether its neighbor uses a slot. Every possible incoming slot is attached to its indicated state there. Only the source cell decides between its state vertex and a fresh leaf, using its own scanned symbol and successor rule. Figure 3 isolates this choice for one slot.

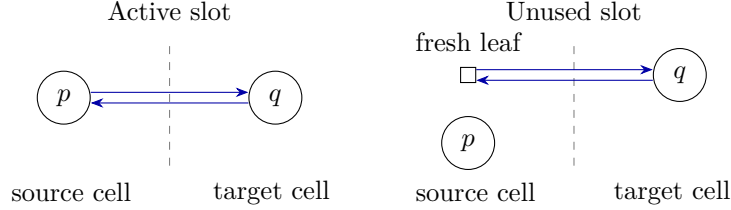


Figure 3: One candidate transition slot with $d = \rightarrow$, with its two travel lanes. The target end is always attached to state q . Only the source attachment changes: an actual transition uses state p , and an unused slot uses its own new leaf, adding no state-to-state connection. Thus the target cell needs no information about the source symbol. Other slots and local cyclic joins are omitted; lane arrows indicate traversal, not two machine transitions.

The two outward boundaries. An ordinary cell has a neighbor on each side. The left-marker cell has only its right neighbor, and the right-marker cell only its left neighbor. Use candidate slots exactly on these inward boundaries. On the outward side of the left marker, instead attach a test edge from c_0 to a new outer leaf z_L . On the outward side of the right marker attach a test edge from t to another new outer leaf z_R .

Each test edge has two lanes whose outer tips occupy ports 1, 2 of the corresponding outward side; the same indices serve as ordinary candidate-lane ports on internal sides. These tips will remain unjoined at the outer leaf. Pair the other outward ports as $(3, 4), (5, 6), \dots, (k-1, k)$, independently of all graph edges. This uses all remaining ports because k is even. Denote the graph including the two test leaves by $\hat{G}$. Its component containing t is a tree, and its two test leaves are connected if and only if the machine accepts.

Lanes and local joins. Place two travel lanes on every edge of $\hat{G}$, one in each direction of traversal. Each incidence consequently has an incoming tip and an outgoing tip. For an edge with incidences i and j , its lanes connect the outgoing tip at i to the incoming tip at j , and the outgoing tip at j to the incoming tip at i . This prescription applies also to a loop, whose incidences are different even though their vertices coincide.

For a crossing edge, each half-lane ends at the corresponding boundary port. Absolute travel directions determine the following local correspondence:

side of the cell	$\rightarrow$ lane	$\leftarrow$ lane
left	incoming tip	outgoing tip
right	outgoing tip	incoming tip

Thus equal-index gluing connects an outgoing half-lane in one cell to an incoming half-lane in its neighbor, as required.

At every vertex other than z_L, z_R , choose a cyclic order of its incidences and join the incoming tip at each incidence to the outgoing tip at the next incidence. A leaf reflects one lane into the other; a vertex of degree zero requires no joins. Choose these orders separately for each symbol type and then use the same orders at every occurrence of that symbol. This is possible because all incidences are local: incoming crossing slots are always present, outgoing attachments depend only on the scanned symbol, and stay edges are internal. Marker incidences, including their test edges, are likewise fixed by marker type. Incidences can, for example, be ordered by their slot labels and their roles in internal edges.

Contracting one cell. Inside a cell, every lane tip is incident with one lane and one local join. The only exposed ends are its boundary ports, each used exactly once, including the separately paired outward ports. Hence the finite wiring inside the cell has degree two at internal points and degree one at its ports. Every component containing a port is a path ending at exactly one other port. Discard components without ports and pair the two endpoints of each remaining path. The result is a perfect matching on the cell's $2k$ ports, hence an element of $\mathcal{D}_k$. Let $\eta(a)$ be this matching for symbol a .

All preceding choices for one cell depend only on a , which proves the asserted symbol locality. Gluing cells with equal port indices reconstructs their full wiring. Contracting paths first within cells or only after gluing yields the same pairing of outside ports, by the definition of diagram multiplication. Consequently the whole-input diagram in the theorem is precisely the pairing obtained from the wires of $\hat{G}$, with the two test-leaf joins omitted and the unused outward ports separately paired. Figure 4 shows the test-port part of this contraction in the accepted case.

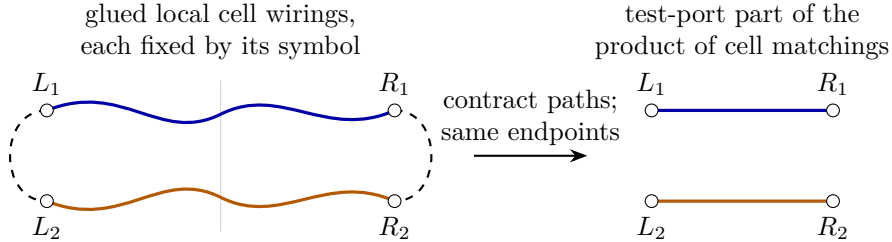


Figure 4: The accepted case, restricted to the four test ports already exposed on the marker outward sides: L_i and R_i name port i at the left and right. For the connected sink tree, Lemma 4.3 shows that restoring the dashed test-leaf reflections closes one tour cycle; opening them leaves two arcs, each joining the two sides. Contracting within cells before gluing or after gluing preserves these outside endpoints and their pairing. The displayed cross pairing is one possibility. Other ports, caps, and wire components of the degree- k construction are omitted.

Reading acceptance. If a wire pairs a left test port to a right test port, it travels only along edges and joins within one component of $\hat{G}$. The separately paired outward ports are disconnected from these wires. Thus z_L and z_R lie in the same component, and the machine accepts.

Conversely, on an accepted input both test leaves lie in the tree component containing t . Temporarily add their degree-one reflection joins. Lemma 4.3 gives a single circular wire containing both joins. Removing them again gives two paths, each from a left test port to a right test port. This proves the acceptance criterion. Other components, including components with cycles, cannot affect these paths.

When $w = \varepsilon$, the input still has two distinct adjacent marker cells. Their inward sides glue by exactly the same slot scheme, and both outward test edges are present. In the connected case the tree has at least these two edges, so the nonempty-tree hypothesis of Lemma 4.3 holds. Thus the argument also gives the criterion for $\eta(\vdash)\eta(\dashv)$.

Finally, under the positive-transition acceptance convention, first perform the following operation on the original machine, before the sink normalization above. Add a new nonaccepting initial state q'_0 with partial row $\delta'(q'_0, b) = \delta(q_0, b)$ for every scanned symbol b , using the original partial transition function. Keep all old states and accepting states, and leave every transition destination in the old state set. Starting at q'_0 simulates exactly the original computation from its first transition onward, so acceptance with the time-zero convention for this $(s + 1)$ -state machine is the desired positive-transition acceptance of the original. Apply the construction already proved to obtain degree $4(s + 2)^2$. $\square$

Extend η to body words by multiplication, with $\eta(\varepsilon) = \mathbf{1}$. If $\eta(u) = \eta(v)$, then for every $x, y \in \Sigma^*$ the whole-input diagrams of xuy and xvy coincide. The theorem therefore makes u and v indistinguishable in all input contexts, including empty contexts.

5 From relation products to the state lower bound

We now connect the two representations. The use of the full relation alphabet makes this step elementary: every relation occurs as a single letter, and two additional letters select any one entry of a product. This is the context-separation property of one-way liveness [1].

Proposition 5.1. *For $h \geq 2$, the language OWL_h has a nondeterministic automaton with $h + 3$ states, with no left moves.*

Proof. Use the h elements of $H = [h]$ as path states, together with three distinct states $q_{\text{in}}, q_{\text{acc}}, q_{\text{rej}}$. Start in q_{in} on the left endmarker and move right into any path state $p \in H$. At a letter $R \in \mathcal{R}_H$ in path state p , move right into any q with $(p, q) \in R$. If there is no such q , enter q_{rej} by a stay move. At the right endmarker in any path state, enter q_{acc} by a stay move. Declare only q_{acc} accepting. The rejecting state stays forever, and all unused rules may enter it by a stay move.

An accepting computation on $R_1 \cdots R_\ell$ specifies states $p_0, \dots, p_\ell$ with $(p_{j-1}, p_j) \in R_j$ for every j . Such states exist exactly when $R_1 \cdots R_\ell \neq 0$. For the empty word, the first right move goes directly from the left marker to the right marker in a path state, and the next step accepts. This agrees with $I_H \neq 0$. All transitions respect the boundaries, all $h + 3$ states are counted, and acceptance always follows a positive transition. $\square$

Lemma 5.2. *Let H be a nonempty finite set. Suppose a deterministic automaton recognizing nonzero relation products over $\mathcal{R}_H$ has a symbol-diagram representation in $\mathcal{D}_k$ as in Theorem 4.1. Let $S \subseteq \mathcal{D}_k$ be the submonoid generated by the ordinary-symbol diagrams, including the empty product. Then there is a unital surjection $S \rightarrow \mathcal{R}_H$ sending each ordinary-symbol diagram to its relation letter.*

Proof. For a body word w , let $d(w) \in S$ be its diagram product and let $r(w) \in \mathcal{R}_H$ be its relation product. Suppose $d(u) = d(v)$. For any body words x, y , associativity gives

$$d_L d(x) d(u) d(y) d_R = d_L d(x) d(v) d(y) d_R,$$

where d_L, d_R are the fixed marker diagrams. The recognition rule therefore gives the same membership decision for xuy and xvy .

If $r(u) \neq r(v)$, choose (i, j) belonging to exactly one of them. Take x to be the single letter $I_{\{i\}}$ and y the single letter $I_{\{j\}}$. For any relation A ,

$$I_{\{i\}}AI_{\{j\}} = \begin{cases} \{(i, j)\}, & (i, j) \in A, \\ 0, & (i, j) \notin A. \end{cases}$$

Exactly one of the two surrounded words would belong to the language, a contradiction. Thus $d(u) = d(v)$ implies $r(u) = r(v)$, and $\phi(d(w)) = r(w)$ is well-defined on S .

Concatenation proves that ϕ preserves multiplication. The empty word gives $\phi(\mathbf{1}) = I_H$, including when a nonempty word also has diagram product $\mathbf{1}$. Every relation is the image of its one-letter diagram, so ϕ is surjective. $\square$

Proof of Theorem 1.1. The source-state assertion is Proposition 5.1. Let an arbitrary s -state deterministic two-way automaton recognize OWL_h on all finite words. Under initial-configuration acceptance, Theorem 4.1 gives a representation in $\mathcal{D}_k$ with $k = 4(s + 1)^2$. Lemma 5.2 supplies a unital surjection from the generated submonoid onto $\mathcal{R}_{[h]}$. Corollary 3.2 then gives

$$4(s + 1)^2 = k \geq L(h) = 2^{\lfloor (h-2)/31 \rfloor}.$$

With positive-transition acceptance, use the additional initial copy in Theorem 4.1; its degree is $4(s + 2)^2$. This proves both stated bounds. $\square$

Remark 5.3. The auxiliary vertices and wires used in the matching representation are an algebraic encoding of a computation graph, not additional memory granted to the deterministic simulator. The lower bound is proved for every machine in the ordinary model stated in the introduction. The family uses a growing finite alphabet, as permitted by the required alphabet-independent constants; the proof makes no separate fixed-alphabet claim.

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